

Juhyeok Park

SENIOR, B.S. ELECTRONIC ENGINEERING, CHUNGBUK NATIONAL UNIVERSITY · EXPECTED FEB. 2027

✉ complimentary656@gmail.com | 🏠 nonoamjam.github.io | 📧 nonoamjam | 🇵🇸 Military service: completed

Research interest Perception and verification: checking an AI's decisions against sensor signals before they are executed, and remembering verified experience to decide faster in physical AI

Education

Chungbuk National University

Cheongju, Korea

B.S. IN ELECTRONIC ENGINEERING

Mar. 2021 – Feb. 2027 (expected)

- GPA 3.80 / 4.5 (major 3.84)
- A+ in Probability and Statistics, Random Processes, Communication Theory, Digital Communication, Digital Signal Processing, Signals and Systems, Information and Communication Engineering

Publications · Patent

- **[First author · oral]** Juhyeok Park et al., “An LLM Wiki System for High-Consequence Domain Knowledge Organization: Initial Feasibility for Space Operations and Nuclear Engineering,” *3rd IAA Conference on AI for Space*, Jeju, Aug. 2026.
- **[First author · oral]** Juhyeok Park et al., “A VLM robot manipulation pipeline for mitigating physical hallucination” (in Korean), *Korea Society of Industrial Information Systems Spring Conference, 2026* (corresponding author: Hogeon Seo).
- **[Co-author]** D. Choi, J. Park, H. Seo, “Development of an AI agent framework for autonomous improvement of an ultrasonic imaging algorithm” (in Korean), *Journal of the Korean Society for Nondestructive Testing*, 46(4), 268–277, 2026 (KCI; corresponding author: Hogeon Seo).
- **[Co-inventor]** “Directional steering in LLM embedding space with a human in the loop,” Korean patent application (pending), 2026.

Research Experience

Korea Atomic Energy Research Institute (KAERI)

Daejeon, Korea

RESEARCH INTERN

Mar. 2026 – Aug. 2026

- Designed and built an LLM wiki knowledge-management system on the nuclear-domain model AtomicGPT: the model writes wiki pages as it works and reads the relevant pages before answering. Retrieval hit rate 55%→79%, answer accuracy 33%→55%. Implemented on internal and external network servers and handed over; now being built into an institute service (IAA 2026, first author)
- Developed a pipeline that checks VLM-generated robot manipulation commands with forward kinematics before execution. Designed controlled experiments over degrees of freedom (1→2→5) and four hint types; collected and analyzed the data and wrote the paper (KSIIIS 2026, first author)
- In a framework where an LLM agent repeatedly edits and evaluates ultrasonic SAFT imaging code, designed the evaluation scheme and benchmark tasks (including an extension for an institute-wide service) and documented the code (J. Korean Soc. NDT 2026, co-author)
- Ran the experiments and built the demo for a patent on directional steering in LLM embedding space (co-inventor)

Capstone: noise-robust object detection in satellite images (advisor: Prof. H. Kim)

Chungbuk National University

TEAM LEAD · MODEL, TRAINING AND EVALUATION CODE

Nov. 2025 – Sep. 2026

- Relational knowledge distillation from a DINOv2 teacher to a YOLOv8s-seg student, combined with noise training. +0.163 mAP50 under noise while keeping the deployed model at 22 MB. Fixed three bugs in the training framework (Ultralytics)
- Built a web demo that compares detections across noise levels

AI Competition on Traffic Accident Analysis from Dashcam Videos (Ministry of the Interior and Safety · National Forensic Service, DACON)

5-PERSON TEAM · STAGE 2 (COLLISION TIME AND APPROACH DIRECTION)

Sep. 2026

- Built models that determine the collision time and the other vehicle's approach direction from dashcam video, and built a labeling tool

NAVIS Lab, Chungbuk National University

Cheongju, Korea

UNDERGRADUATE RESEARCHER

Dec. 2025 – Feb. 2026

- Designed and implemented two compensation algorithms for estimating the arrival time of eLoran signals under sample loss on top of the baseline code, and validated them with 1,000 Monte Carlo runs and measured eLoran signals. SNR improved by 6.78 dB but biased the arrival time; plain accumulation proved the most robust

Coursework · Skills

Courses

Electronic Circuits · Digital System Design · SoC Design · Computer Architecture · AI Systems · AI System Semiconductor Design · Automatic Control

Course projects

PPM modulation/demodulation with hard and soft decisions (MATLAB) · Cortex-M3 secure OTP · AES-128/CBC from scratch · Raspberry Pi real-time vision and voice integration

Languages · Tools

Python, C/C++, MATLAB · PyTorch, Ultralytics YOLO, DINOv2 · LLM/VLM pipelines, built on-premise LLM service servers · Linux, Git, $\LaTeX$

Papers, slides and demos: nonoamjam.github.io